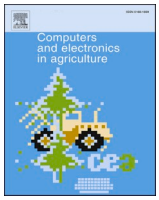
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Optimizing potato yield mapping and prediction: Integrating satellite-based remote sensing and machine learning for sustainable agriculture

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ABSTRACT

Precision agriculture and sustainable farming require crop yield prediction and mapping. PEI is a major Canadian potato producer. However, PEI potato yield prediction research is limited. This highlights a literature gap and the need for improved data-driven precision agriculture in PEI. High-resolution satellite imagery and machine learning (ML) enable field-scale crop yield mapping. This study investigated the potential of high-resolution multispectral imagery and ML for potato yield prediction. The study focused on four plots in PEI during the 2021 and 2022 growing seasons. Potato crop yield data collected using a combined harvester and manual digging were analyzed to model yield using Sentinel-2A and PlanetScope imagery. For both sensors, vegetation indices (NDVI, GNDVI, SAVI, and EVI) and spectral bands chosen for their application in crop growth monitoring were retrieved and incorporated into ML models. The cloud computing platform, Google Earth Engine (GEE), was used to evaluate and compare the performance of three ML algorithms, which are random forest regression (RFR), classification and regression trees (CART) and gradient tree boosting (GTB). Overall, the performance of all three models was satisfactory in yield prediction with both sensors. However, GTB with Sentinel-2A and harvester data gave slightly higher estimation accuracy with R^2 values of 0.71–0.78, RMSE values of 2.82–5.96 t/ha, and MAE values of 2.33–4.2 t/ha. Hence, the approach used in this study provides real-time seasonal yield prediction maps on a field scale, which will help the farmers identify the targeted areas for variable rate application, leading to resource efficiency and sustainability.

1. Introduction

Potato cultivation is an essential element of the global food chain, serving millions worldwide as a fundamental food supply. In 2021, Canada was listed in the top twelve countries for potato production, yielding approximately 41.3 t/ha yearly from over 150,000 ha of land (FAOSTAT, 2021). Prince Edward Island (PEI) ranks as Canada's third-largest potato producer, contributing 20.4 % (S. Canada, 2024). It is recognized for its fertile, iron-rich red soils, which are optimal for potato production. Potatoes are cultivated in the spring and summer (May to October), primarily relying on rainfed conditions. Potatoes are vulnerable to climate change, presenting a significant concern for growers in PEI (Canada, 2024). Consequently, in response to climate change, accurate high-resolution potato yield maps are beneficial for potato crop production management, elucidating regional variability patterns and significant elements affecting yield while offering insights into management strategies (Bian et al., 2022; Li et al., 2021).

The traditional methods of potato yield prediction are based on visual assessments, historical data analysis, pest and disease monitoring, and empirical and mathematical crop growth models. Although these techniques are often inexpensive and valuable, they may not be as precise as recent tools and methods. Moreover, the computation of yield using traditional methods is no longer beneficial for decision-makers because it takes excessive time (Singha & Swain, 2023). The

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advancement of innovative information technology has recently facilitated the development of remote sensing techniques in crop yield prediction, including next-generation satellites, aerial and ground sensors, and ML algorithms (Lin et al., 2023). The use of remote sensing technology has greatly improved the accuracy of crop yield predictions by providing a reliable source of comprehensive, multi-temporal vegetation data (Zhang et al., 2025).

Many studies have shown that integrating remote sensing and ML algorithms improves predictive precision, scalability, and decision-making abilities relative to conventional approaches (Bharadiya et al., 2023). Singha and Swain (2023) assessed the support vector machine (SVM), random forest (RF), logistic regression, and Naïve Bayes models for estimating rice and potato crop yields utilizing vegetative variables, including NDVI, DVI, PVI, RVI, SAVI, NDWI, BI, GEMI, and LAI, derived from Sentinel-2B. Their findings revealed that RF, SVM, and logistic regression attained an accuracy of 80–90 %, whilst naïve Bayes was categorized as a moderate to decent model with an accuracy of 60–80 %. Similarly, another study done by Faqe Ibrahim et al. (2023) used biophysical parameters of the canopy, including Leaf Area Index (LAI), and biochemical properties, such as chlorophyll, along with spectral vegetation indices (VIs) (NDVI, EVI, CMFI, RVI, SAVI, NDWI, SR, NDRI and GRVI), utilizing Google Earth Engine (GEE), derived from Sentinel-2 and Landsat 8, to forecast wheat yield via a linear regression model. The results revealed that Sentinel-2 performed better with an RMSE of 0.56 and an R^2 value of 0.77 than Landsat 8 with an R^2 of 0.71 and an RMSE of 1.54. Prasad et al. (2021) implemented an RFR machine learning algorithm to forecast cotton yield in three phases before harvest in Maharashtra, India. The RF model exhibited dependable performance in September, December, and February, attaining R^2 values of 69 %, 60 %, and 39 % for yield predictions. CART decision trees and recursive feature elimination were employed to identify critical variables.

The results emphasized the RF algorithm's capacity to incorporate a wide range of satellite and ground-based data while preserving precision and preventing overfitting. Jhajharia and Mathur (2023) utilized several ML algorithms, including random forests, decision trees, support vector regression and lasso regression, on vegetation indices obtained from remote sensing and meteorological data to forecast crop yield. The RF model achieved an RMSE of 0.39 t/ha, R^2 of 0.77, and a mean absolute error of 0.28 t/ha, indicating remarkable predictive accuracy.

The choice of remote sensing parameters is essential for accurate yield mapping. The structural, spectral, textural and thermal properties of crop canopies are crucial indicators of yield variability (Maimaitijiang et al., 2020). Vegetation indices derived from two or more bands are essential tools that represent crops' biophysical and biochemical properties. They facilitate information capture due to their robust relationship with crop output at different developmental phases. The most frequently employed VI in agriculture for predicting potato yield is the normalized difference vegetation index (NDVI). In addition, numerous researchers recommend and implement the enhanced vegetation index (EVI), green normalized vegetation index (GNDVI), and soil adjusted vegetation index (SAVI) for precise potato yield mapping (Abou Ali et al., 2020; Lin et al., 2023; Mukiibi et al., 2024).

The literature reviewed above indicates that using satellite-based remote sensing data (vegetation indices) and ML algorithms to predict and map yields with high accuracy will improve crop productivity and ensure food's consistent and reliable production, ensuring food availability for a growing population. Our literature review indicates that research on estimating potato crop productivity in PEI is sparse, with only a few studies utilizing the ML model (Abbas et al., 2020). However, satellite-based remote sensing data has yet to be investigated, creating a notable void in the literature for this region.

This study assesses the efficacy of PlanetScope and Sentinel-2A imagery to estimate potato yield using the cloud-based GEE platform (Fig. 1). Four potato plots were chosen in the 2021 and 2022 crop seasons, and three machine-learning models were evaluated for each plot. This study aims to (a) compare the spatial and temporal resolutions of PlanetScope and Sentinel-2A for field-scale potato yield prediction and mapping; (b) evaluate the accuracy of different machine learning models in predicting potato yield using multispectral satellite data; (c) examine the efficiency of ground-based yield data collection methods, such as harvester measurements and manual digging. This study's findings will give farmers early warnings on potential yield loss, enabling them to enhance farm profitability through data-driven practices.

The remainder of the manuscript is organised as follows: Section 2 provides an overview of the study area, the datasets used and presents the adopted methodology, including data preprocessing and machine learning models. Section 3 presents the experimental results and analysis. Section 4 presents and discusses the obtained results, their implications and directions for future work. Finally, Section 5 concludes the study.

2. Materials and methods

2.1. Study area

This research examined four sampling sites cultivated with potatoes over two years in Prince Edward Island (PEI), Canada (Fig. 2). Plots A

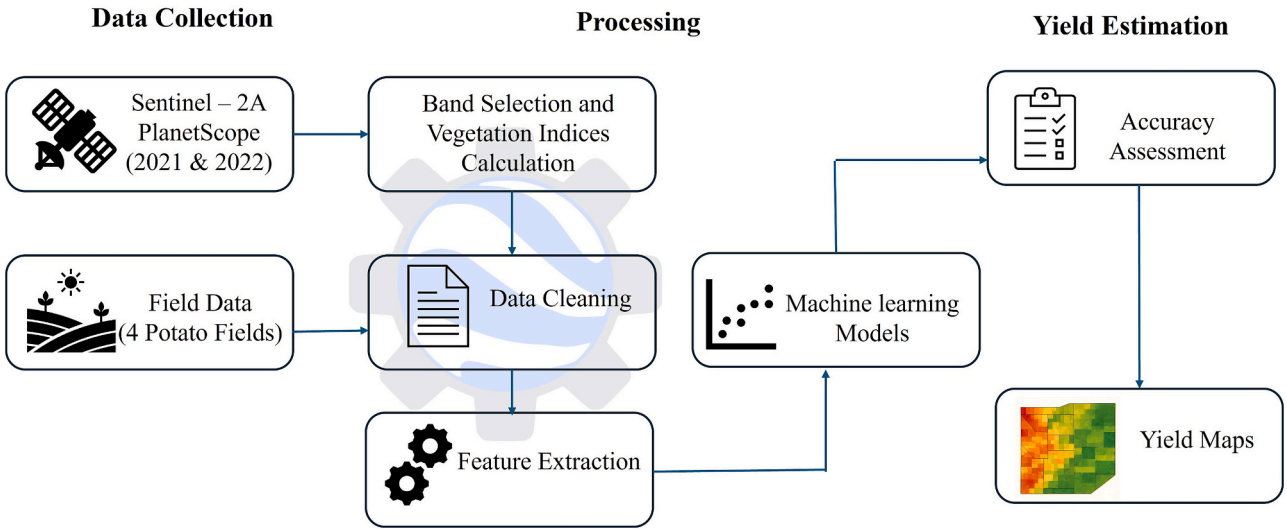


Fig. 1. Overall architectural diagram illustrating the key stages of the proposed approach, including data collection, processing, and yield estimation.

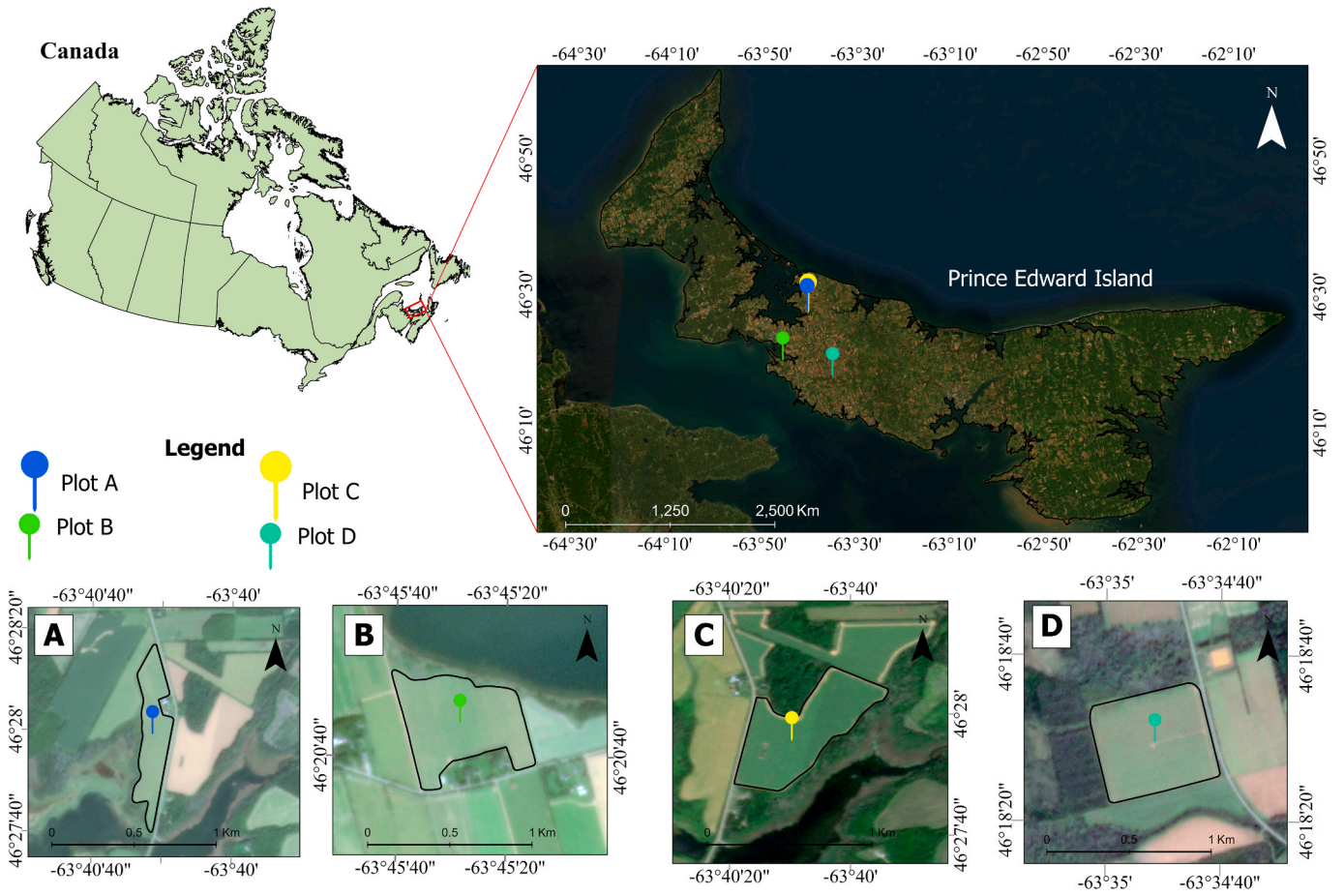


Fig. 2. Study area with field scale layout maps for the observed plots (A and B) during the 2021 crop season and (C and D) during the 2022 crop season. The RGB true color composite was obtained from PlanetScope.

(46.46443°N, −63.67309°W) and B (46.34527°N, −63.75822°W) were observed for the 2021 crop season, and Plots C (46.46543°N, −63.66936°W) and D (46.30817°N, −63.58096°W) were studied in the 2022 crop season. All four plots cover an approximate area of 15–16 ha. The potatoes were planted at the end of May and harvested in early October. Rainfed agriculture is a common practice in PEI. The average annual precipitation in the study area is approximately 890 mm, with an average temperature of −8 °C in January and 19 °C in July (Weather, 2024).

2.2. Remote sensing data

2.2.1. Sentinel 2-A

Multispectral Instrument data from Sentinel-2 (Level 2A) (accessible at https://developers.google.com/earthengine/datasets/catalog/COPERNICUS_S2_SR_HARMONIZED) (retrieved on 14 November 2024) was obtained directly via GEE. Level 2A products are already pre-processed for atmospheric corrections. The spatial resolution for bands 2, 3, 4, and 8 is 10 m; for bands 5, 6, 7, 8a, 11, and 12 is 20 m; and for bands 1, 9, and 10 is 60 m. This study employed red, green, blue and Near-infrared bands (Table 1) of Sentinel-2A satellite imagery, characterized by minimal cloud cover (< 10 %) from June to September throughout two farming seasons of 2021 (five images) and 2022 (7 images), utilizing the available capabilities on the GEE platform.

Moreover, the QA60, which includes bitwise flags for the cloud contamination band of Sentinel-2A, has been utilized to mask clouds and Cirrus. Only the pixels that were devoid of cloud cover or cirrus clouds were retained. To enhance data quality, residual artifacts were further reduced using temporal compositing, utilizing mean values across dates (Agency 2023).

Table 1
Spatial details of sentinel 2-A bands utilized in this study.

Band number	Band name	Central wavelength (λ_o)/ Full width at half maximum (FWHM) (nm)	Spatial resolution (m)
2	Blue	490 /65	10
3	Green	560/35	10
4	Red	665/25	10
8	Near-infrared	842/115	10

2.2.2. PlanetScope

This study utilized PlanetScope's cloud-free (no clouds and cloud shadows) daily 4-band imaging from the PSB.SD instrument at a resolution of 3 m (Table 2) was retrieved directly from the Planet Explorer

Table 2
Spatial details of PlanetScope bands utilized in this study.

Band number	Band name	Central Wavelength (λ_o)/ Full width at half maximum (FWHM) (nm)	Spatial resolution (m)
1	Blue	455/40	3
2	Green	545/40	3
3	Red	630/50	3
4	Near-infrared	820/40	3

webpage (https://www.planet.com/explorer/, accessed on September 25, 2022). The imagery was downloaded for two agricultural seasons (June to September) in 2021(17 images) and 2022 (19 images). The imagery was subsequently uploaded to GEE for further processing as an image collection via the Assets tab.

2.3. Field data collection and cleaning

Two types of yield data were used in this study: (a) yield data collected using a GPS-enabled harvester and (b) yield data collected by a manual dig through field sampling.

For plots A and C, the potato yield data were obtained using an Allens GPS-enabled combine harvester with yield monitoring technology. After collecting the monitoring data, they were cleaned, edited, and transformed into yield maps. Shapefiles (.shp) containing the raw yield data were exported, and further data cleaning was done in ArcGIS Pro using the attribute table. First, all the points with zero yields were removed. The data was further cleaned for outliers using crop-dependent thresholds and Inter-Quartile range (Lyle et al., 2014; Sanchez et al., 2023).

For plots B and D, each field was divided into three zones, and the potato yield data were obtained through manual harvesting of a 10 ft strip at 30-point locations, which were evenly spread out in the field, having 10 points in each zone. Yields for the point samples were determined by weighing the cleaned tubers on a battery-operated balance (Fig. 3), calculating the yield, and converting it into the standard unit (t/ha) (Khan et al., 2021).

To match the spatial resolution of the satellite data, inverse distance weighted interpolation was applied to the point vector layers of all four observed plots to convert them into 3 and 10 m raster grids corresponding to the spatial resolutions of PlanetScope and Sentinel-2A, respectively.

Finally, to maintain the sample size across all four plots, 300 points were randomly generated in each plot using ArcGIS Pro, and the yield values were extracted from these points to use this field data for prediction. This data was split into 70 % for model training and 30 % for validation.

2.4. Vegetation indices (VIs) calculation

In this study, according to the literature, four different vegetation indices (VIs), including the Soil-Adjusted Vegetation Index (SAVI), Normalized Difference Vegetation Index (NDVI), Enhanced Vegetation Index (EVI), and Green Normalized Difference Vegetation Index (GNDVI), derived from satellite data, were used. The vegetation indices were calculated for PlanetScope and Sentinel-2A in GEE. For Sentinel-2A, an image collection was generated by selecting a product (with the product ID: "COPERNICUS/S2_SR_HARMONIZED) and filtering it for the crop season; for both sensors, to create the composite image, image reduction was applied using the mean function. The vegetation indices were computed using the bands given in Tables 1 and 2 on the 300 randomly generated points in ArcGIS Pro and distributed evenly inside the potato field boundaries.

Based on the literature (Luo et al., 2019; Singha & Swain, 2023), these indices are estimated as follows:

$$NDVI = \frac{NIR - Red}{NIR + Red} \tag{1}$$

$$GNDVI = \frac{NIR - Green}{NIR + Green} \tag{2}$$

$$EVI = \frac{2.5(NIR - Red)}{NIR + 6 \times Red - 7 \times Blue + 1} \tag{3}$$

$$SAVI = \frac{(NIR - Red)}{(NIR + Red + L)} \times (1 + L) \tag{4}$$

Where:

NIR: Near infrared band.
Blue: Blue band.
Green: Green band.
Red: Red band.
L: 0.5.

The soil adjustment factor (L) in the SAVI formula was set to 0.5, following the recommendation by Singha and Swain (2023).

2.5. Correlation analysis between vegetation indices and yield

After calculating the vegetation indices, a correlation analysis was done to look at the relationship between potato yield and the vegetation indices produced from Sentinel-2A and PlanetScope images before model building. The linear associations among the variables were assessed using Pearson correlation coefficients (Fig. 4). Along with NIR, NDVI, SAVI and GNDVI, the red, green, blue, and NIR bands were also used as independent variables in the model, while the dependent variable was actual potato yield, derived from ground-truth data. The selection of these variables was based on their accepted significance in assessing crop vigour, canopy structure, and chlorophyll content, all of which exhibit a substantial correlation with yield results.

2.6. Machine learning (ML) models

This study has incorporated three ML models: Random Forest Regression (RFR), an ensemble-based algorithm; Classification and Regression Tree (CART), a single tree method; and Gradient Tree Boosting (GTB), which is a combination of both an ensemble and a tree-based method. All three models were implemented in GEE. These



Fig. 3. Manual harvesting (A) and weighing (B) of potatoes.

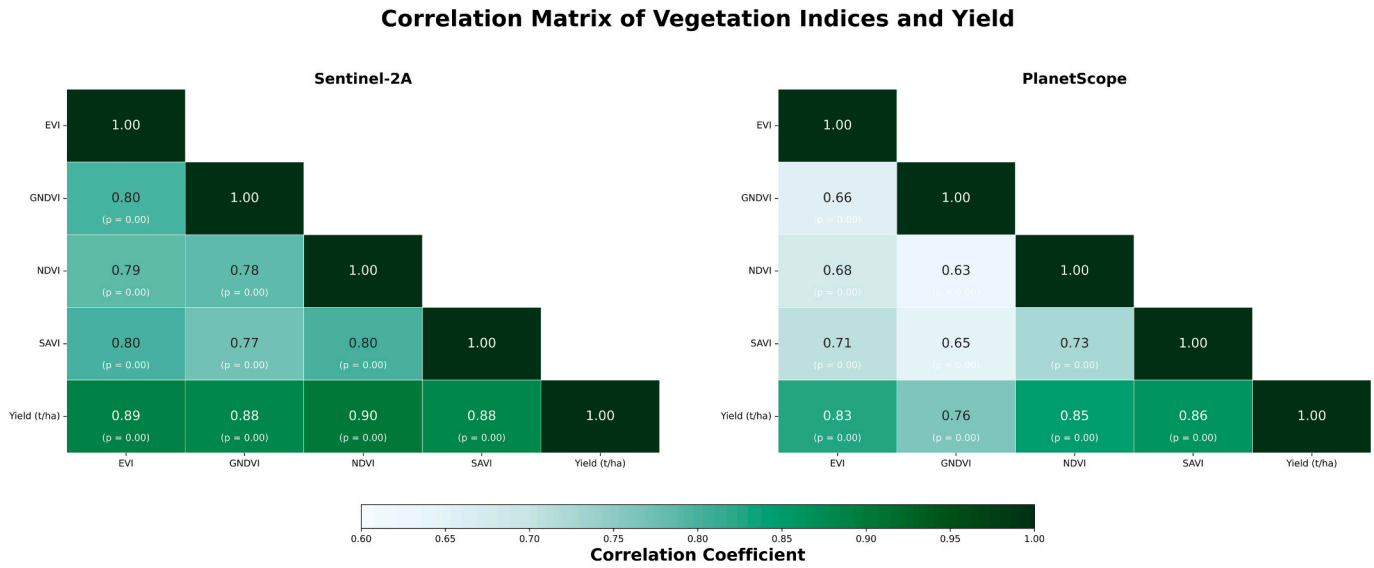


Fig. 4. Correlation heatmaps between various vegetation indices and potato crop yield.

models were picked based on the demonstrated robustness and performance in attaining high predictive accuracy for potato yield in previous studies (Gómez et al., 2019; El-Kenawy et al., 2024). In particular, RFR and GTB have consistently ranked among the most frequently used and accurate ML methods in crop yield prediction literature (Shawon et al., 2024; Mukiibi et al., 2024).

RFR builds multiple decision trees for regression and classification methods. It is a smart, versatile, and robust model that drives the prediction by averaging the outputs of all regression trees (Breiman, 2001). CART operates by iteratively splitting the training data into progressively smaller subsets and creating a single tree, and the predictive value is the mean of the target variable (Savitha & Talari, 2025). Using a decision tree as a starting point, GTB iteratively creates new models by fixing the mistakes in the original weak learner. This iterative approach decreases prediction residual errors and increases the model's accuracy. GTB is well-known for its capacity to identify intricate relationships in data and for producing predictions with a high degree of accuracy (Gaikwad et al., 2024).

To prevent cross-field data leakage, each of the four plots was modelled independently. Using just the data from each plot, the machine learning models were trained and validated with a 70/30 random split between training and validation sets. This method guaranteed independent evaluation of model predictions for every potato plot and made sensor-wise comparisons (Sentinel-2A and PlanetScope) inside reasonable spatial limits possible.

Hyperparameters were individually tweaked for each field and satellite sensor via exploratory experimentation (trial and error process). As there are no built-in, automated hyperparameter optimization tools within GEE, its capabilities have some limitations relative to dedicated ML platforms, and the procedure is more manual and less scalable (Ríos-Mesa et al., 2024; Savitha & Talari, 2025). Hence, we analysed performance on the validation set and chose combinations that delivered the optimum balance of accuracy and generalisability. Table 3 shows the specific parameters utilised for each field and sensor.

2.7. Accuracy assessment

The coefficient of determination (R^2), root mean square error (RMSE) and mean absolute error (MAE) were utilized to evaluate the correlation between predicted and observed parameters, as indicated in equations (5) – (7). These performance indicators were chosen based on their extensive use in previous literature for assessing model validity and prediction performance (Hoque et al., 2024; Anand & Jhajharia, 2025). Models exhibiting the highest R^2 and the lowest RMSE (t/ha) and MAE (t/ha) were found. To compare the predictive performance of candidate regression models, a bootstrap resampling approach with B = 1000 iterations was applied (Champagne et al., 2014). In each iteration, models were trained on a bootstrap sample and evaluated on the out-of-bag data. Performance metrics RMSE and R^2 were recorded across all iterations. The mean values and 95 % confidence intervals (CIs) were calculated to assess model accuracy and stability. The model with the lowest RMSE and highest R^2, along with narrow CIs, was selected as the best predictor.

$$R^2 = 1 - \frac{\sum_{i=1}^{n} \left(y_i^m - y_i^p\right)^2}{\sum_{i=1}^{n} \left(y_i^m - y_i^a\right)^2} \quad (5)$$

Table 3
Hyperparameters considered for ML models for potato yield prediction. Tested hyperparameter values. maxNodes. 100, 120, 150; nTrees. 100,150,200; minLeafPop. 1, 3, 5, 7; shrinkage. 0.1, 0.07, 0.05.

Sensor	Plot	CART (maxNodes)	RF nTrees	minLeafPop	GTB nTrees	shrinkage
Sentinel-2A	A	100	150	3	100	0.1
	B	100	200	1	100	0.1
	C	150	100	5	150	0.05
	D	100	200	1	100	0.1
PlanetScope	A	100	150	3	100	0.05
	B	100	200	3	100	0.1
	C	120	150	7	150	0.07
	D	100	200	1	100	0.1

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(y_i^m - y_i^p\right)^2} \quad (6)$$

$$MAE = \frac{1}{n}\sum_{i=1}^{n}\left|\left(y_i^m - y_i^p\right)\right| \quad (7)$$

The actual potato yield data (t/ha) of the ith sample is represented by y_i^m. The predicted potato tuber yield data (t/ha) of the ith sample is represented by y_i^p in the yield estimation model. The mean value of the actual potato marketable yield (t /ha) dataset is denoted by y_i^a. The sample size is represented by n, and i denotes the sample index.

2.8. Flowchart of study

Fig. 5 illustrates the flowchart of this study. It consists of four significant steps: (a) input data, (b) Pre-processing, (c) Analysis and (d) Results.

3. Results

3.1. Performance evaluation of ML models

This study examined the applicability and accuracy of three machine-learning models utilizing two distinct machine-learning algorithms for predicting potato yield in PEI. A field-scale machine learning model utilizing high-resolution Sentinel-2A (10 m) and PlanetScope (3 m) imagery was developed to estimate seasonal potato yield across four plots monitored during the 2021 and 2022 crop seasons. The yield monitor data from these plots was used to validate these models. Each model's performance was evaluated over all four plots in GEE to determine the optimal model.

Figs. 6 and 7 depict the scatterplots from the correlation analysis between the observed and expected yields, evaluating the performance

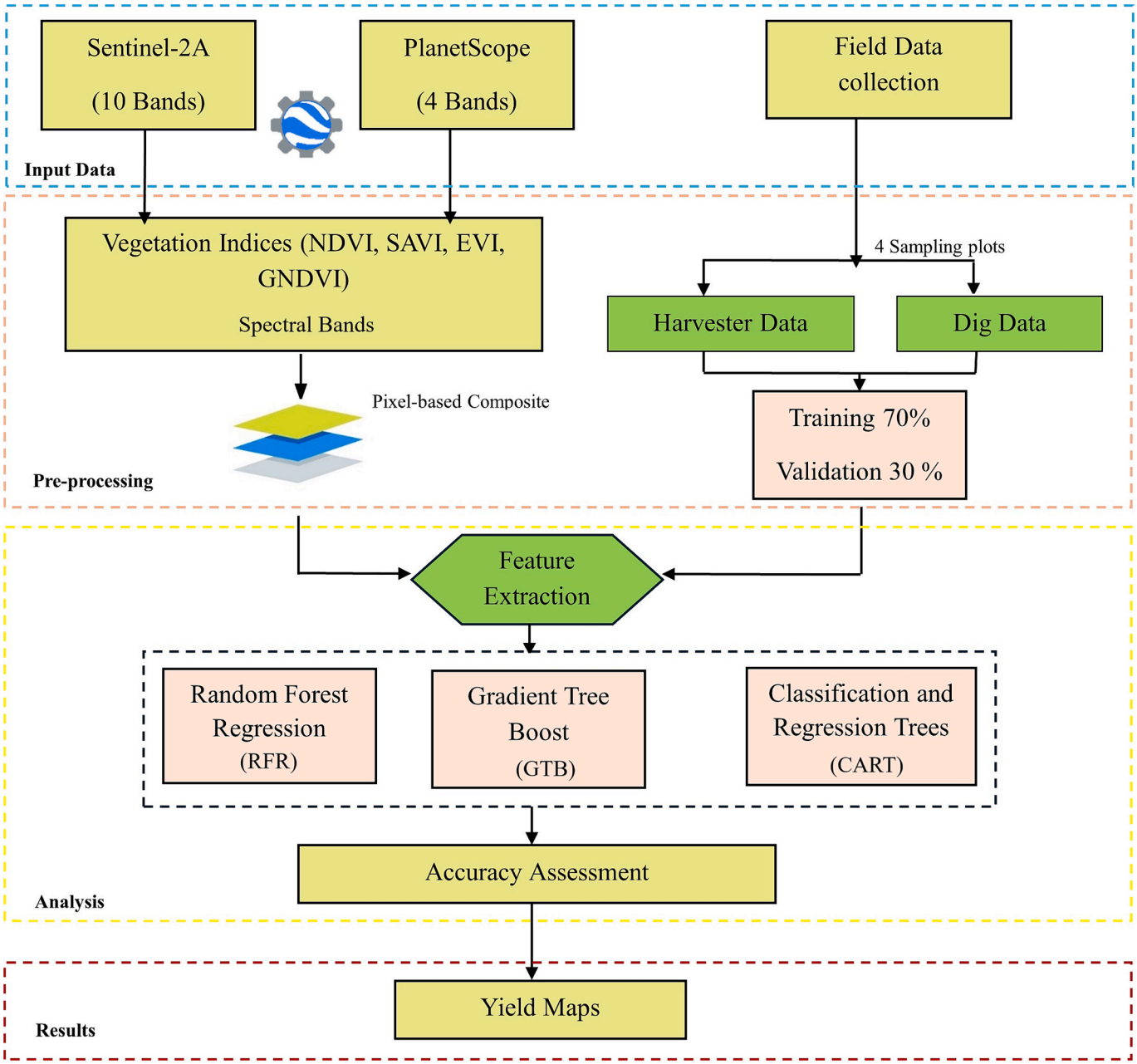


Fig. 5. Flow chart of the implemented methodology for Potato yield prediction for the 2021 and 2022 crop seasons.

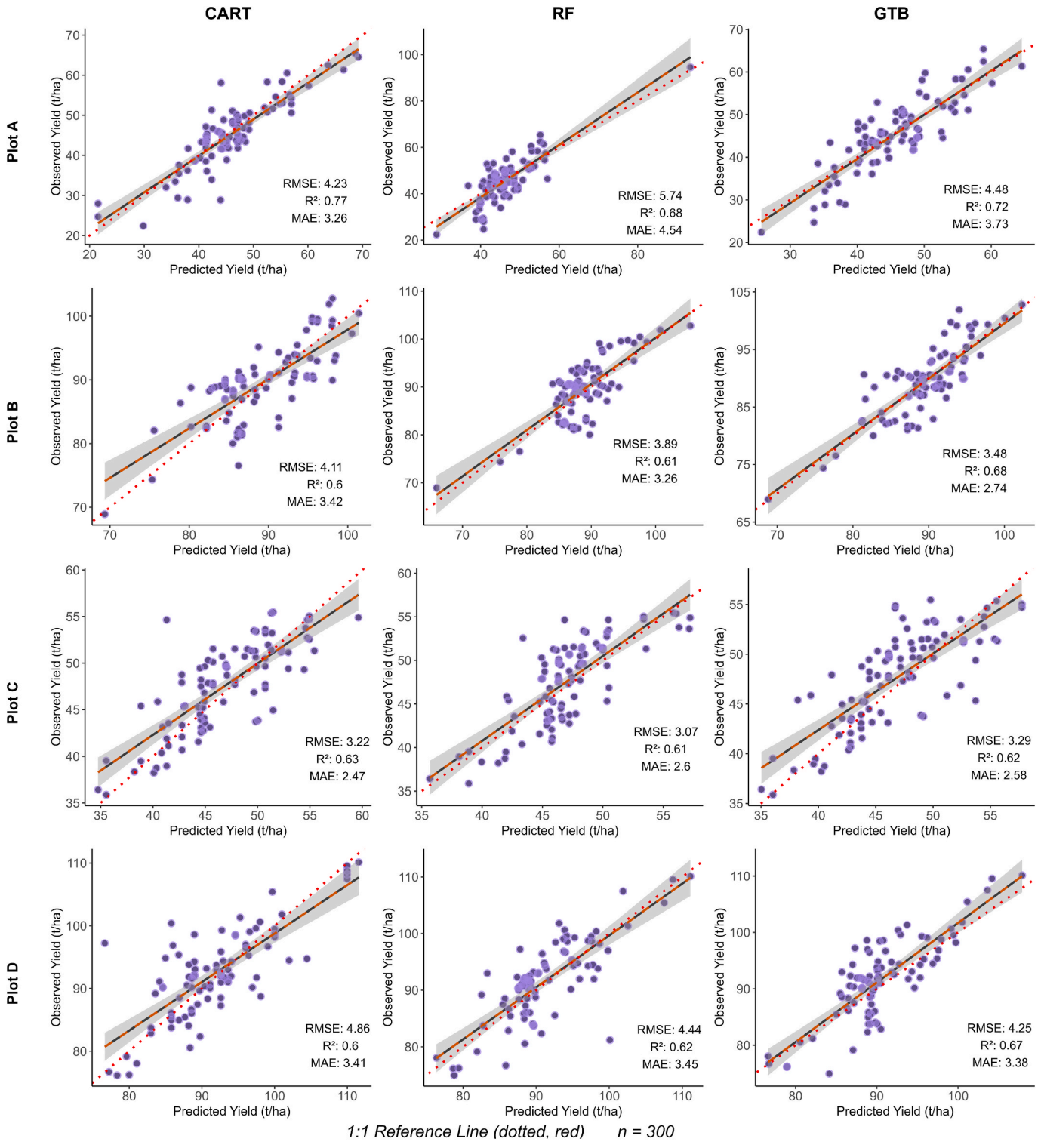


Fig. 6. Scatter plot of the correlation between the observed and predicted potato yields of all four plots with CART, RFR and GTB models using Sentinel-2A spectral bands and vegetation indices.

and accuracy of each model utilized by PlanetScope and Sentinel-2A spectral bands and vegetation indices. Tables 4 and 5 show the performance metrics of ML models over all four plots. Based on the results, GTB model attained the best prediction accuracy with both satellites, exhibiting R^2 values of 0.62–0.78, RMSE values of 2.82–5.96 t/ha, and MAE values of 2.33–4.2 t/ha using Sentinel-2A data, and R^2 values of 0.67–0.72, RMSE values of 3.29–4.48 t/ha, and MAE values of 2.58–3.73 t/ha using PlanetScope data. CART demonstrated commendable estimation accuracy, yielding R^2 values between 0.6 and 0.76, RMSE ranging from 2.92 to 4.57 t/ha, and MAE between 2.5 and 3.47 t/ha when utilizing Sentinel-2A data, as well as R^2 values from 0.6 to 0.77, RMSE from 3.22 to 4.86 t/ha, and MAE between 2.47 and 3.42 t/ha with PlanetScope data. However, the prediction accuracy of RFR is somewhat inferior to that of GTB and CART, exhibiting R^2 values between 0.61 and 0.72, RMSE values ranging from 2.94 to 5.79 t/ha, and MAE values from 2.47 to 4.07 t/ha when utilizing Sentinel-2A data. For

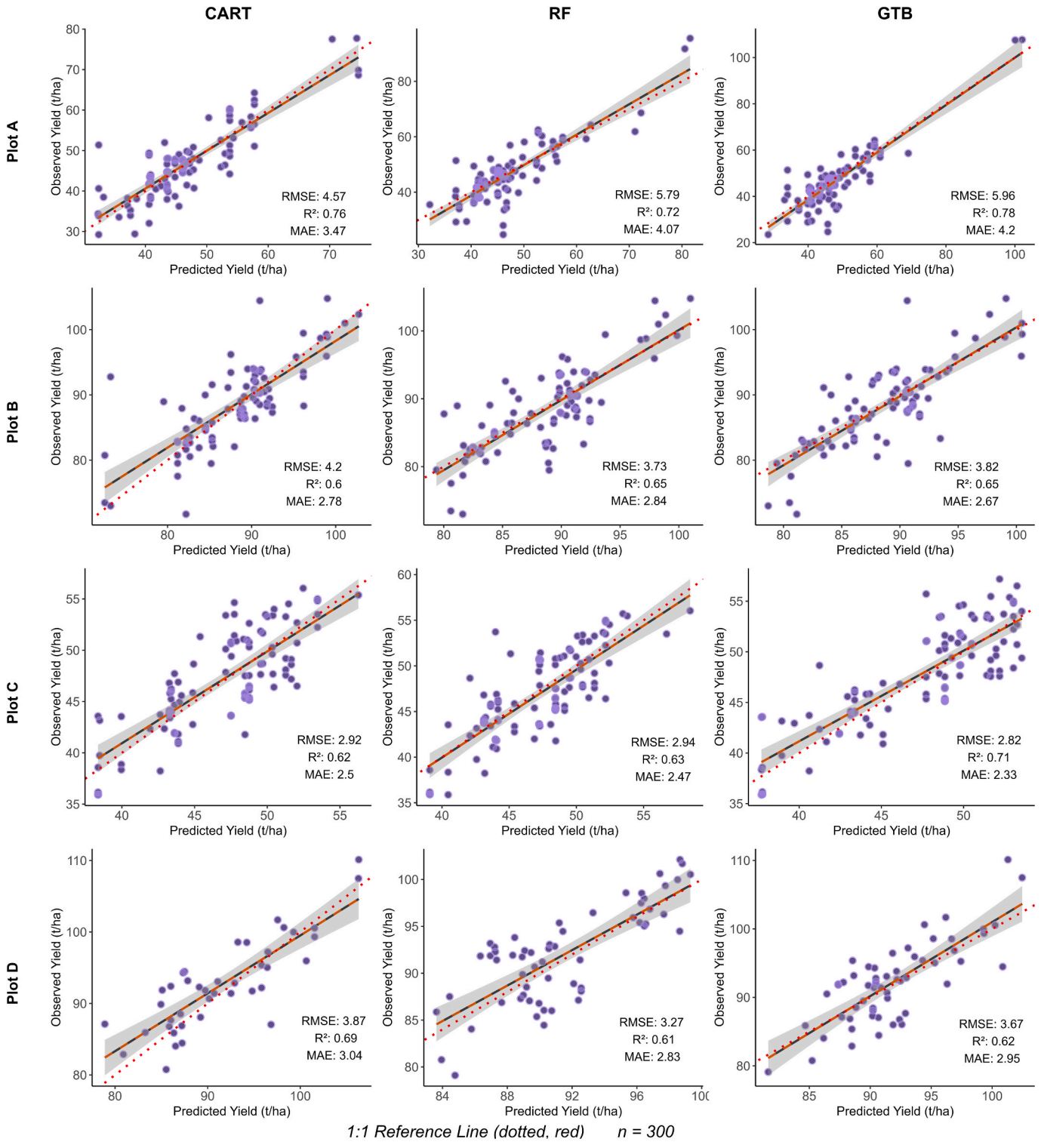


Fig. 7. Scatter plot of the correlation between the observed and predicted potato yields of all four plots with CART, RF and GTB models using PlanetScope spectral bands, and vegetation indices.

PlanetScope data, R^2 values are between 0.61 and 0.68, RMSE values range from 3.07 to 5.74 t/ha, and MAE values range from 2.6 to 4.54 t/ha (Tables 3 and 4). Based on the results presented in Table 6 using the bootstrap resampling approach, the GTB, CART, and RFR models demonstrated the best predictive performance, as indicated by their lower RMSE, higher R^2 values, and narrower 95 % confidence intervals. In summary, GTB and CART outperform RFR in predicting potato yield across all four plots for the 2021 and 2022 crop growth seasons. The best-performing model was used to create field-scale maps to facilitate the subsequent analysis.

3.2. *Sentinel-2A vs PlanetScope*

This study investigated the prospect of utilizing multispectral satellite data to assess potato yield at the field level. Based on the high-performance accuracy of GTB, we further compared PlanetScope and

Table 4
Performance metrics of the applied ML models using sentinel-2A bands and VIs for potato yield prediction.

	Model	Training RMSE (t/ha)	MAE (t/ha)	R^2	Testing RMSE (t/ha)	MAE (t/ha)	R^2
Plot A	CART	4.2	3.2	0.82	4.57	3.47	0.76
	RF	5.3	3.7	0.78	5.79	4.07	0.72
	GTB	5.5	3.8	0.84	5.96	4.2	0.78
Plot B	CART	3.9	2.4	0.67	4.2	2.78	0.6
	RF	3.4	2.5	0.72	3.73	2.84	0.65
	GTB	3.5	2.3	0.72	3.82	2.67	0.65
Plot C	CART	2.6	2.1	0.7	2.92	2.5	0.62
	RF	2.6	2.1	0.7	2.94	2.47	0.63
	GTB	2.5	2	0.78	2.82	2.33	0.71
Plot D	CART	3.4	2.6	0.75	3.87	3.04	0.69
	RF	2.9	2.4	0.68	3.27	2.83	0.61
	GTB	3.2	2.5	0.7	3.67	2.95	0.62

Table 5
Performance metrics of the applied ML models using PlanetScope bands and VIs for potato yield prediction.

	Model	Training RMSE (t/ha)	MAE (t/ha)	R^2	Testing RMSE (t/ha)	MAE (t/ha)	R^2
Plot A	CART	3.9	3	0.82	4.23	3.26	0.77
	RF	5.2	4	0.75	5.74	4.54	0.68
	GTB	4	3.2	0.78	4.48	3.73	0.72
Plot B	CART	3.7	2.9	0.66	4.11	3.42	0.6
	RF	3.5	2.7	0.68	3.89	3.26	0.61
	GTB	3.2	2.4	0.74	3.48	2.74	0.68
Plot C	CART	2.9	2.1	0.69	3.22	2.47	0.63
	RF	2.8	2	0.67	3.07	2.6	0.61
	GTB	3	2	0.7	3.29	2.58	0.62
Plot D	CART	4.5	3	0.74	4.86	3.41	0.69
	RF	4	2.9	0.7	4.44	3.45	0.62
	GTB	3.9	2.8	0.72	4.25	3.38	0.67

Sentinel-2A data, considering the differences between the sensors and the balance between cost and accuracy. The results in Fig. 8 showed that the Sentinel-2A-based GTB model predicted the yield with somewhat higher accuracy (RMSE = 2.82–5.96 t/ha, R^2 = 0.62–0.78 and MAE = 2.3–4.2 t/ha) than the high-resolution PlanetScope-based model (RMSE = 3.29–4.48 t/ha, R^2 = 0.62–0.72 and MAE = 2.58–3.73).

3.3. Harvester vs manual dig

The two methods for yield data collection (harvester data and manual dig data) utilized in this study were compared based on the GTB model to estimate potato yield at the field scale. The results in Fig. 9 showed that the yield data collected using the combine harvester (Plots A and C) has higher accuracy than the manual dig data (Plots B and D). For Sentinel-2A, the R^2 values were 0.78 for Plot A and 0.71 for Plot C. The RMSE values were 5.96 for Plot A and 2.82 for Plot C, whereas the MAE values were 4.2 for Plot A and 2.33 for Plot C. The R^2 values for PlanetScope were 0.72 for Plot A and 0.62 for Plot C. The RMSE values were 4.48 for Plot A and 3.29 for Plot C, whereas the MAE values were 3.73 for Plot A and 2.58 for Plot C.

3.4. Field-scale spatial yield estimation and mapping

The optimum performing GTB model was employed to delineate the pixel-level spatial distribution of crop yield over all four fields, as illustrated in Fig. 10. The red rectangles have been included on the predicted yield maps to indicate the discrepancies in predictions based on satellite data. For Plot A, the performance metrics indicate the minimal quantitative difference (Table 7) between the two sensors; a slight visual inconsistency was observed near the field border of Sentinel-2A, which was absent in PlanetScope, attributed to the high-resolution images of PlanetScope occasionally encountering the edge effect.

In Plot B, the predicted yield map from Sentinel-2A exhibited a smoother and more homogeneous appearance. In contrast, the red rectangle that marks the corresponding area in PlanetScope displayed a salt-and-pepper effect, likely attributable to the smaller size (3 m) of PlanetScope, where fine-grained variations appeared as noise. In Plot C, the spatial pattern in the predicted yield map differs for each sensor, reflecting the variations in the metric values. The spatial pattern of PlanetScope appears more homogeneous, likely due to its elevated R^2 value. Additionally, the difference in spatial patterns from both sensors arose from the number of images used in the analysis for each sensor and the difference in acquisition dates. Overall, the results from Sentinel-2A appear more realistic, considering both the spatial patterns and the model's performance metrics.

4. Discussion

The study analyzed the performance of remote-sensing-derived multispectral bands and vegetation indices for field-scale yield prediction employing several machine-learning methods. To accomplish this, we utilized data from high-resolution Sentinel-2A and PlanetScope. Both sensors used red, green, blue, and NIR bands and four vegetation indicators (NDVI, EVI, GNDVI, and SAVI) as predictor variables for yield prediction. Furthermore, yield monitor data from the harvester and manual digging were used. The effectiveness of three ML models, Gradient Tree Boosting (GTB), Classification and Regression Trees (CART), and Random Forest Regression (RFR), was assessed. These models' accuracy was evaluated using three performance metrics: R^2, RMSE and MAE.

The overall accuracy of all models was satisfactory in the order of GTB, CART, and RFR. However, according to the evaluation metrics for all the observed plots, GTB was the best model.

These results are similar to those of other studies done on yield

Table 6
Performance metrics of the applied machine learning models with bootstrapped 95 % confidence intervals for R^2 and RMSE.

	Model	RMSE t/ha (Mean)	RMSE t/ha (Lower)	RMSE t/ha (Upper)	R^2 (Mean)	R^2 (Lower)	R^2 (Upper)
Plot A	CART	7.85	5.49	10.30	0.58	0.41	0.66
	GTB	5.89	4.76	7.04	0.74	0.43	0.88
	RF	7.10	5.46	8.74	0.64	0.43	0.75
Plot B	CART	4.34	3.23	5.52	0.52	0.20	0.75
	GTB	3.81	3.02	4.58	0.64	0.49	0.77
	RF	3.96	3.26	4.62	0.61	0.42	0.74
Plot C	CART	3.33	2.81	3.84	0.55	0.36	0.70
	GTB	2.82	2.44	3.20	0.68	0.55	0.78
	RF	3.47	2.97	4.02	0.52	0.31	0.67
Plot D	CART	4.57	3.46	5.67	0.39	0.26	0.75
	GTB	3.72	3.04	4.40	0.61	0.39	0.74
	RF	4.46	3.50	5.45	0.45	0.17	0.61

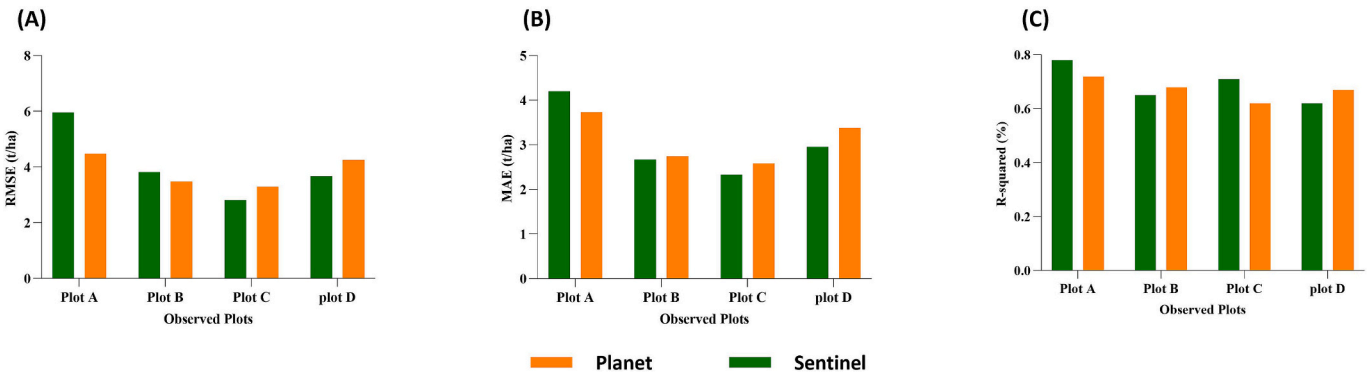


Fig. 8. Comparison of Sentinel-2A vs PlanetScope based on performance metrics of the GTB model.

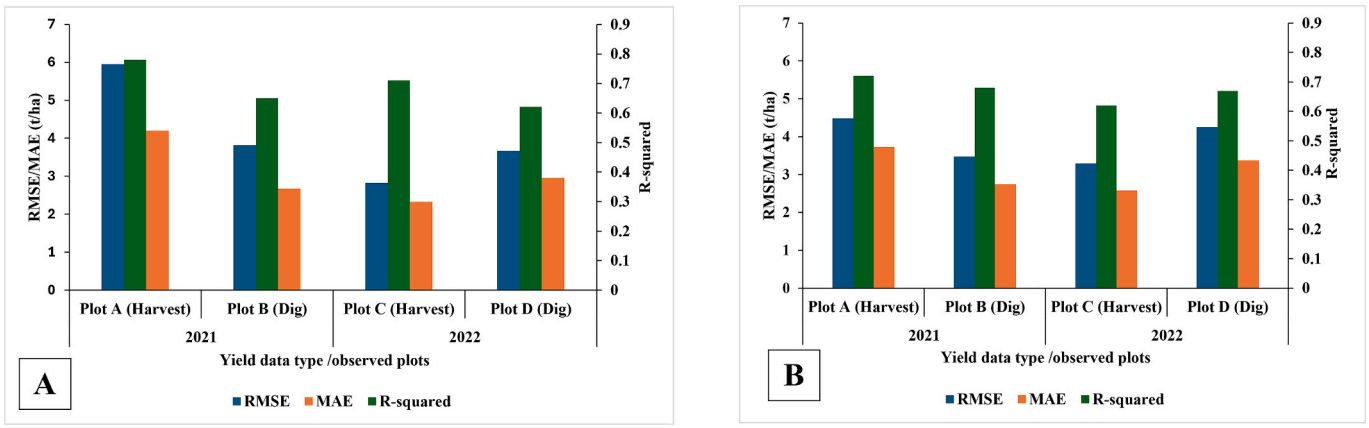


Fig. 9. Comparison of harvester vs dig data based on the performance metrics of the GTB model. A-Sentinel-2A and B-PlanetScope.

prediction. For Example, in a study by Jahromi et al. (2023), they predicted wheat yield using satellite-based vegetation indices and meteorological variables, including evapotranspiration. They found that the gradient tree boosting algorithm provided slightly more accurate yield predictions than random forest algorithms. Similarly, another study done by Du et al., 2022 for corn yield prediction using soil organic matter and weather data gradient tree boosting decision tree (R^2 = 0.799) showed better accuracy than the random forest model (R^2 = 0.749). A study done in Australia for field-level cane and sugar yield prediction using Sentinel-1 and Sentinel-2-based vegetation indices also revealed that gradient tree boosting was the most accurate method for predicting field-scale yield with an R^2 of 0.63 and a mapping accuracy of 73.4 %. The high accuracy of GTB is attributable to its more sensitive nature to outliers relative to other models. Furthermore, GTB integrates numerous decision trees (weak learners) to formulate a robust predictor, frequently yielding exceptional performance (Shendryk et al., 2021).

Sentinel-2A outperformed PlanetScope based on the performance evaluation of the GTB model in all four plots, and we got better accuracies using the combined harvester data. Similar results were reported by Farmonov et al. (2023), who compared Sentinel-2A and PlanetScope for wheat yield prediction using nine vegetation indices and spectral bands and found that Sentinel-2A performed slightly better than PlanetScope. Despite PlanetScope's excellent spatial and temporal resolution, the validation results in the current study revealed that PlanetScope bands are not always better at explaining potato yield variability at the field scale. Even though the pre-processing, some noise may remain in the surface reflectance bands of PlanetScope, which can be more noticeable in small fields (Liu et al., 2022). The residual errors, inter-sensor discrepancies, and the radiometric normalization process may also contribute to this minor variation in results from Sentinel-2A (Houborg & McCabe, 2018). Similarly, Frazier and Hemingway (2021) assert that Planet's foundational processing (geometric and radiometric quality) is inadequate for intricate analytical procedures.

Furthermore, the temporal discrepancies between satellite and reference data and the possible influence of varying surface units also compromise data accuracy. However, the temporal resolution of the PlanetScope provides the opportunity for higher accuracies, which can be achieved by some data fusion techniques of both sensors, as done by Kumar et al. (2023). In addition, despite the low spatial and temporal resolution of Sentinel-2A, we can achieve comparable results to PlanetScope in terms of its better calibration (Li et al., 2020).

The potential of remote sensing and machine learning for high-resolution mapping of potato yields on PEI was explored in this study, producing reliable results through the use of multispectral bands and vegetation indices. However, it is crucial to acknowledge the study's limitations. First, the likelihood of spatial autocorrelation within the training and validation datasets may result in overestimating model performance. Second, the model's ability to represent within-season variability in crop development phases was restricted due to the absence of detailed phenological data caused by the use of single seasonal composites. As sometimes, even with cloud masking and low cloud cover, this limitation persists since unfavourable weather continues to impact image quality and temporal consistency. Thirdly, the training data may contain errors or biases resulting from the limited size and inadequate spatial distribution of the ground-truth dataset, requiring interpolation in case of manual dig data. Fourth, to ensure the uniformity of PlanetScope data with Sentinel-2A, only comparable bands were used, excluding the red-edge and SWIR bands of Sentinel-2A, which are essential for showing water condition and vegetation structure. Finally, the use of commercial PlanetScope imagery reduces repeatability due to financial or licensing constraints on high-resolution datasets. By using larger and more evenly distributed ground datasets, adding more

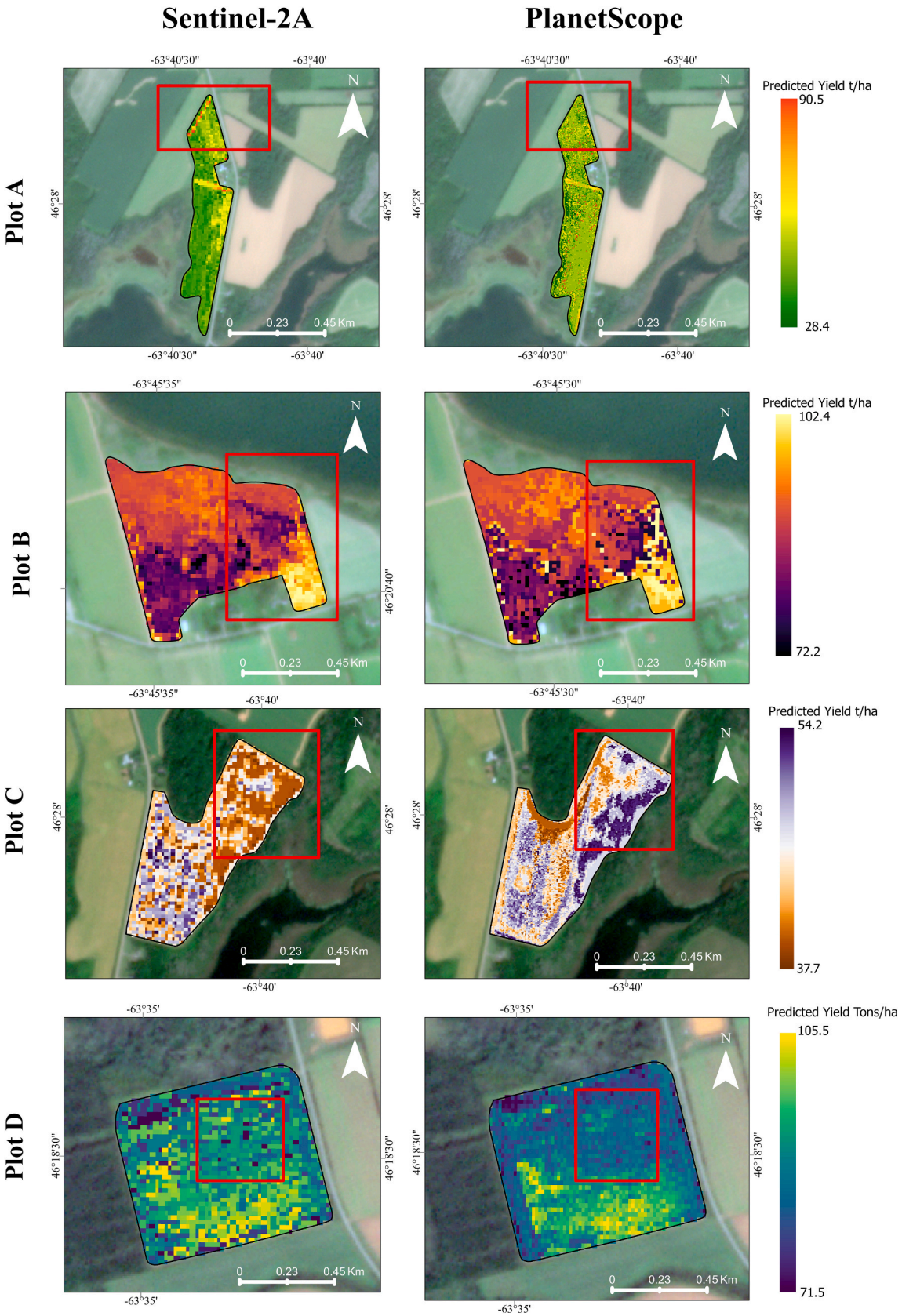


Fig. 10. Field-scale maps of potato yield prediction with the GTB model for the plots observed in 2021(Plots A and B) and 2022 (Plots C and D).

Table 7
Difference in performance metrics of sentinel-2A and PlanetScope potato yield prediction using GTB model.

	Metric	Sentinel-2A	PlanetScope	Difference
Plot A	R^2	0.78	0.72	0.06
	MAE (t/ha)	4.2	3.73	0.47
	RMSE (t/ha)	5.96	4.48	1.48
Plot B	R^2	0.65	0.68	−0.03
	MAE (t/ha)	2.67	2.74	−0.07
	RMSE (t/ha)	3.82	3.48	0.34
Plot C	R^2	0.71	0.62	0.09
	MAE (t/ha)	2.33	2.58	−0.25
	RMSE (t/ha)	2.82	3.29	−0.47
Plot D	R^2	0.62	0.67	−0.05
	MAE (t/ha)	2.95	3.38	−0.43
	RMSE (t/ha)	3.67	4.25	−0.58

spectral bands, and leveraging open-access images, prediction models can be made more generalizable and stable. These limitations should be addressed in future research.

Furthermore, crop production depends on various aspects, such as climate, weather conditions, soil properties, fertilizer use, and seed variety (Xu et al., 2019). To enhance the robustness and generalizability of the current methodology, future studies should focus on testing this methodology using the combination of different input datasets (environmental and bio-physical) along with vegetation indices and more advanced ML models with the integration of radiative transfer models (RTM) ((Alam et al., 2024) and deep learning algorithms (Joshi et al., 2023; Mohan et al., 2023). Additionally, the incorporation of Synthetic Aperture Radar (SAR) and multi-temporal composites offers a great solution to mitigate the limitations associated with cloud cover. Lastly, multiple crop varieties must be tested in this location to evaluate the effectiveness of the current method on a larger scale.

5. Conclusion

This study used satellite imagery from PlanetScope and Sentinel-2A, as well as three machine learning algorithms (RFR, CART, and GTB), to forecast and map potato yields across four plots on PEI, Canada. The primary conclusions of the study are summarised as follows:

The model was trained and validated using yield monitoring data from harvester sensors and manual digging. The models were tested over two crop seasons (2021 and 2022) with four vegetation indices (NDVI, GNDVI, SAVI, EVI) and four spectral bands (red, green, blue, and NIR).

Among the evaluated algorithms, GTB continually outperformed RFR and CART, reaching the highest predictive accuracy. Sentinel-2A data yielded marginally superior results compared to PlanetScope, with R^2 values between 0.71 and 0.78, RMSE values from 2.82 to 5.96 t/ha, and MAE values ranging from 2.33 to 4.2 t/ha.

This study shows that combining multispectral satellite imagery with machine learning may accurately forecast spatially detailed yields. The research sheds light on field-scale production variability, with practical implications for applying precision agricultural practices in PEI and similar agroecosystems.

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CRediT authorship contribution statement

Fatima Imtiaz: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Formal analysis, Data curation. **Aitazaz A. Farooque:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization. **Gurjit S. Randhawa:** Writing – review & editing, Validation, Supervision, Investigation, Funding acquisition. **Xiuquan Wang:** Writing – review & editing, Investigation, Methodology, Validation. **Travis J. Esau:** Writing – review & editing, Formal analysis, Software, Visualization. **Seyyed Ebrahim Hashemi Garmdareh:** Writing – review & editing, Formal analysis, Software, Validation. **Bishnu Acharya:** Writing – review & editing, Validation, Investigation, Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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